MetaCyc — 代谢通路数据库

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简介

MetaCyc 全称 MetaCyc Metabolic Pathway Database，属于 BioCyc 子数据库。

- MetaCyc 是一个由生物学家基于大量的科学文献手工注释的、具有经过实验验证报道的代谢途径和酶的数据库， 确保了高质量和高准确性。
- MetaCyc 包含参与初级和次级代谢的通路，以及相关的化合物、反应、酶和基因。
- MetaCyc 的目标是通过存储每个实验阐明的通路的代表性样本来对代谢领域进行编目。

> BioCyc：
>
> - BioCyc 是由 SRI International（SRI）创建，可以通过 Ovid 平台访问的，以生物体为中心的数据库集合，包含基因组、代谢途径和其他相关生物信息，并配有一套可以进行相关的搜索，浏览，分析和可视化处理的软件工具。
> - 截至到 2023 年 12 月 8 日（27.5 版），BioCyc 包含了 20,050 个数据库，其中第一层数据库包括： MetaCyc (多种生物代谢途径)，HumanCyc (人类代谢途径), EcoCyc (大肠杆菌 Escherichia coli K-12), AraCyc（拟南芥）, YeastCyc（酿酒酵母）。第二层、第三层数据库详情请查看 BioCyc PGDB list。
> - BioCyc 涵盖超过2万个生物体，内容来自超过14万种出版物，提供百科全书式的数据，可用于生物信息学，遗传学，生物医学工程，代谢组学和传染病相关的研究，适用于本科、研究生和更高水平的研究人员和从业者，也可用于大规模计算研究、转录组学、代谢组学和比较数据分析、定量建模、以及在制药和医疗保健，化妆品，生物技术和食品生产中的合成生物学。

> BioCyc 数据库详细介绍

数据库详情

MetaCyc 数据库统计数据

版本 28.0

数据类型	数量
Genes	15,525
Pathways	3,538
Enzymatic Reactions	19,017
Transport Reactions	980
Polypeptides	17,336
Protein Complexes	4,877
Enzymes	13,902
Transporters	685
Compounds	19,196
tRNAs	8
Growth Media	19
Protein Features	32,149
GO Terms	85,012

与 KEGG 数据库的比较

关于 MetaCyc 与 KEGG 的全面比较在 2013 年[2]就有发表.

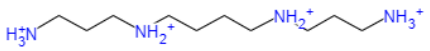
一些比较列举:

- 相较于 KEGG, MetaCyc 提供的代谢物信息更多，除了基础的物质信息以外，还包括 物质的化学性质 （如：油水分配系数、拓扑极性表面积、标准吉布斯自由能等）；在线搜索示例:

compound
spermine
MetaCyc

Add to SmartTable
Provide Feedback

Chemical Formula	$C_{10}H_{30}N_4$
Molecular Weight	206.37 Daltons
Monoisotopic Mass	202.2157468554 Daltons



Summary | Ontology | Reactions | Regulation | References | Structure | Show All

BioCyc Id	SPERMINE
Synonyms	gerontine musculamine neuridine diaminopropyl-tetramethylenediamine
SMILES	C(CCC[NH2+]CCC[NH3+])[NH2+]CCC[NH3+]
InChI	InChI=1S/C10H26N4/c11-5-3-9-13-7-1-2-8-14-10-4-6-12/h13-14H,1-12H2/p+4
InChIKey	InChIKey=PFNFFQXMRSDOHW-UHFFFAOYSA-R
Log P	-1.1
Topological Polar Surface Area	88.5 Å
Standard Gibbs Free Energy of Formation ($\Delta_f G^\circ$)	318.74066 kcal/mol [Latendresse13]
Component of	spermine phosphate

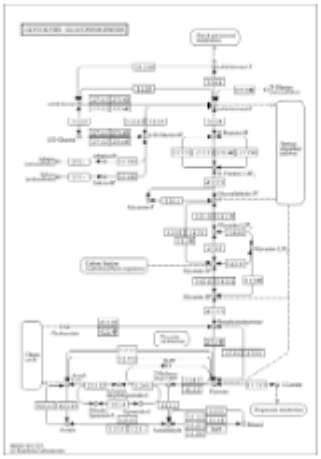
Unification Links

CAS	71-44-3
ChEBI	45725
ChemSpider	394003
DrugBank	DB02564
HMDB	HMDB01256
Kegg	C00750
MetaboLights	MTBLC45725
MetaNetX	MNXM408
PubChem	446718
RefMet	Spermine
Seed	cpd00558
ZINC	ZINC000001532734

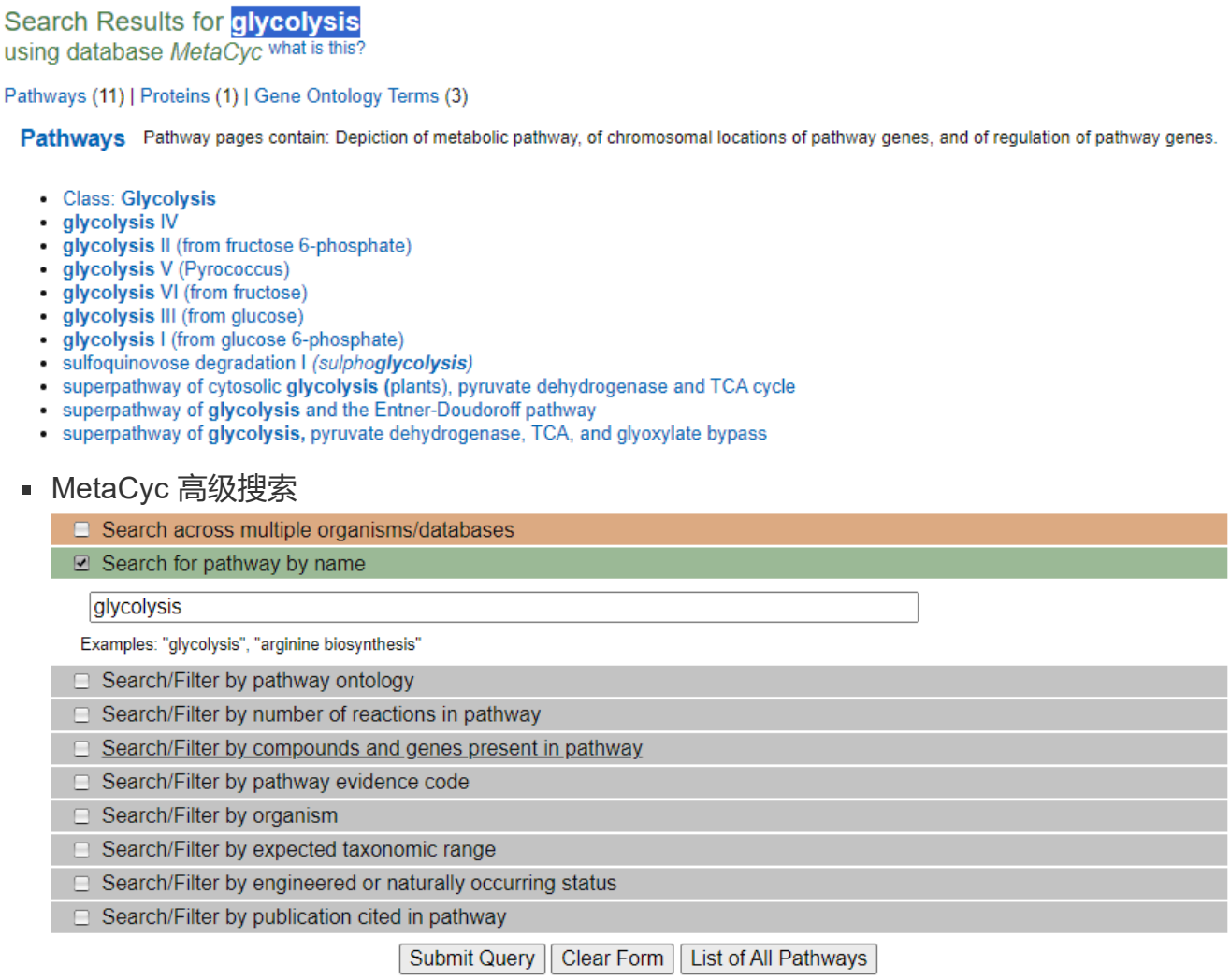
Summary

The history of spermine biochemistry goes back more than 300 years. When Antoni van Leeuwenhoek looked at human semen through the lenses of his microscope in 1678, he noted the formation of crystals in aging sperm. More than 200 years later, the basic component of these phosphate crystals was named spermine, after the source [Galston90].

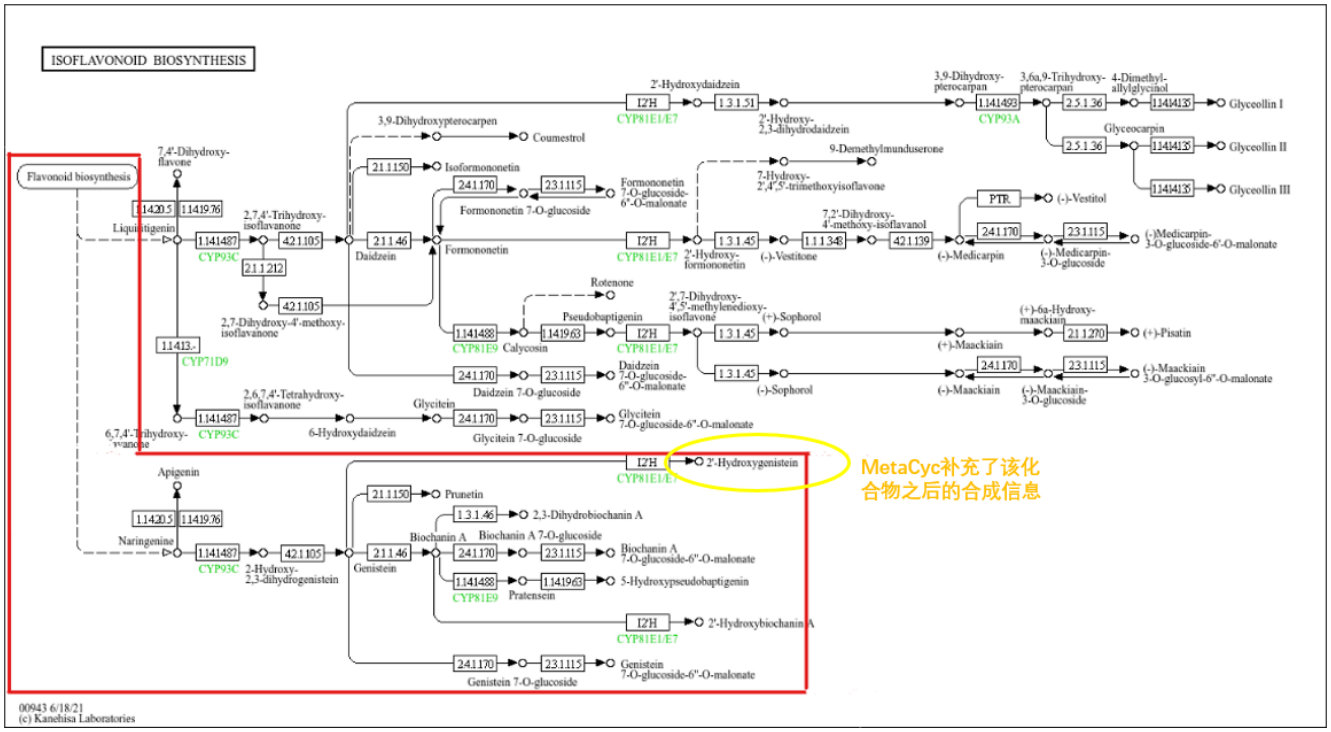
- KEGG 在通路方面的检索方式比 MetaCyc 更简单一些，通过通路名称或一个代谢物即可检索到相关的通路，而 MetaCyc 除了通路名称外，还需要提供其他信息进行筛选，也就是需要 高级搜索 功能才可以锁定一个通路；
  在线搜索示例（以 `glycolysis` 为例）：
  - KEGG

Entry	Thumbnail Image	Name
map00010		Glycolysis / Gluconeogenesis

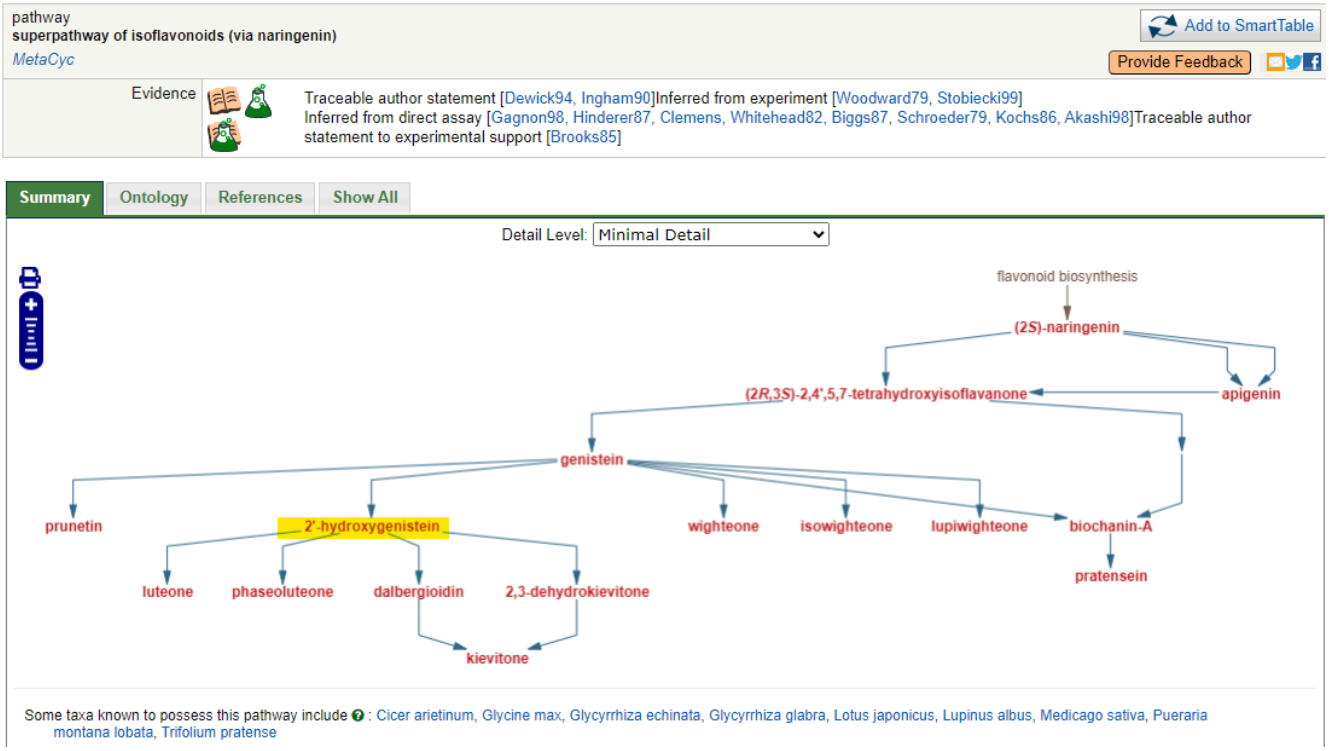
  - MetaCyc，会搜索出11个通路名称中包含'glycolysis'的通路



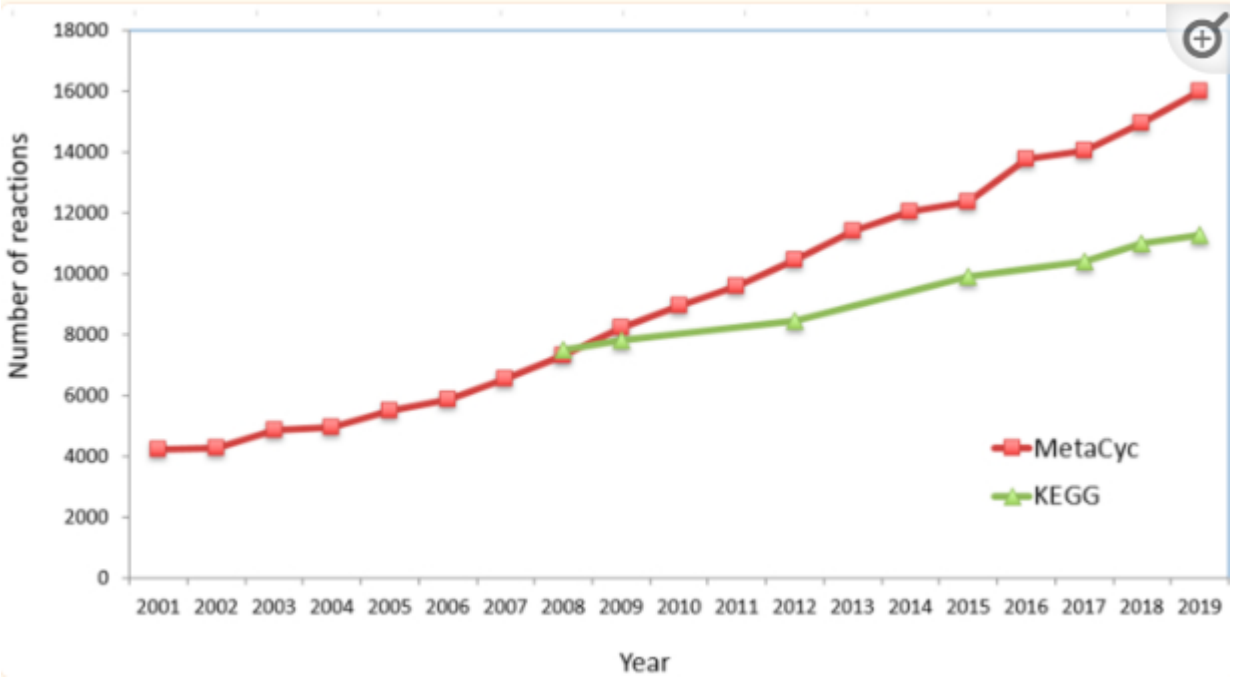
- 另外，在通路的完整度上，KEGG 中更加注重的是在所有物种中的通路汇总到一张图上，而 MetaCyc 更加注重的是不同物种中通路的差异化。
- KEGG 部分通路会覆盖的更全一些，而 MetaCyc 相对会少一些，但是 MetaCyc 可以补充部分 KEGG 通路上不全的部分，因此 KEGG 与 MetaCyc 可以 相互补充 ，达到相得益彰的效果。
  在线搜索示例（以 `isoflavonoids` 为例）：
  - KEGG



- MetaCyc, 补充了 `2'-hydroxygenistein` 之后的合成信息



- MetaCyc 与 KEGG 数据库中 `reaction` 增长趋势对比[1]:



MetaCyc 数据下载

根据文献[1]中对 MetaCyc 数据库获取的描述，截止到2017年都是免费获取，之后因资金问题改为订阅：

SUBSCRIPTION MODEL FOR BIOCYC ACCESS AND DATABASE CURATION Go to:

In past papers in the database issue of *Nucleic Acids Research* we described the MetaCyc database together with the BioCyc PGDB collection (6). As of 2017, SRI International has adopted a subscription-based model for BioCyc access, preventing its description in the database issue. MetaCyc and EcoCyc [the PGDB for *Escherichia coli* K-12, (22)] remain freely available to all and do not require a subscription.

Because of ongoing difficulties in securing government funds for database curation, we moved to a subscription-based model in the hope of generating funds that would permit us to curate high-quality databases for more organisms, such as important pathogens, biotechnology workhorses, model organisms and promising hosts for biofuels development. In the last 2 years, we used subscription revenues to improve the quality of the PGDBs of the following organisms: *Bacillus subtilis* 168, *Corynebacterium glutamicum* ATCC 13032, *Homo sapiens, Lactobacillus rhamnosus* GG, *Mycobacterium tuberculosis* H37Rv, *Pseudomonas putida* KT2440, *Saccharomyces cerevisiae* S288c, *Salmonella enterica enterica* 14028S, *Salmonella enterica enterica* LT2 and *Staphylococcus aureus* NCTC 8325.

More information about the subscription model is available at http://www.phoenixbioinformatics.org/biocyc/index.html.

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Pricing Level	For-Profit/Corporate 2023 (USD)
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Level 5	$28,768
Level 4	$20,537
Level 3	$14,363
Level 2	$9,517
Level 1	$4,897

MetaCyc 相关文献

[1] Caspi20: Caspi R, Billington R, Keseler IM, Kothari A, Krummenacker M, Midford PE, Ong WK, Paley S, Subhraveti P, Karp PD (2020). "The MetaCyc database of metabolic pathways and enzymes - a 2019 update." Nucleic Acids Res 48(D1);D445-D453. PubMed

[2] Altman T., Travers M., Kothari A., Caspi R., Karp P.D.. A systematic comparison of the MetaCyc and KEGG pathway databases. BMC Bioinformatics. 2013; 14:112. PubMed